

Is Your Organisation's Data Secure?

A company's data is priceless and must be resilient to security threats. Fortunately, there are a host of open source tools that can ensure it's protected from breaches.



Millions of records containing customer password information, patient medical records, and customer financial information are continuously being leaked. A common cause is that the data was stored in plain text format or transmitted over unencrypted communications. Had this data been encrypted, the attacker could not have accessed it. Encryption converts readable data (i.e., plaintext data) into an unreadable format (e.g., ciphertext) that can be converted back to the original state only by using a valid cryptographic key.

Historically, strong encryption was seen as costly, complex, and only available to larger corporations that had a dedicated IT security team and

a commercial software licence. This is no longer true. Many free, open source encryption tools are being used around the world and outnumber the proprietary tools in this domain. Examples include OpenSSL, which provides SSL for a significant percentage of HTTPS internet traffic today, and LUKS, which is included in nearly every Linux distribution. Let's Encrypt has issued millions of free TLS/SSL certificates since 2014. The fact that the source code is public makes it possible for cryptographers from many parts of the globe to evaluate it continuously, and vulnerabilities are identified sooner compared to closed systems.

Regulatory forces further support the encryption of personal and financial data, either mandating or strongly

encouraging it in accordance with frameworks such as the General Data Protection Regulation (GDPR), Health Insurance Portability and Accountability Act (HIPAA), Payment Card Industry Data Security Standard (PCI DSS), and India's Digital Personal Data Protection Act of 2022.

Understanding the two states of data

Data at rest: Information kept permanently is known as 'data in storage'. Data in storage can be found in hard disk drives (HDD), solid state drives (SSD), database files (DB), backup tapes (BT), object storage, and removable devices (e.g., USB). Attackers are highly motivated to attack such data because it provides